

PAPER

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Stereoisomers of metal–organic dimers induced by the steric constraint of dissymmetric terpyridine ligands†

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Herein, we report a novel molecular design strategy for constructing stereoisomers based on dissymmetric modified ligands and closely packed dimeric structures. X-ray analyses reveal that different substituents have an important effect on stereoisomers and can provide various driving forces to realize the adjustment of crystal packing.

Stereoisomerism is one of the most fundamental and indispensable laws in nature and plays vital roles in various fields, including natural product synthesis,^{1,2} photochemistry,^{3,4} and catalysis.^{5,6} The understanding of stereoisomerism in biological macromolecules (proteins, nucleic acids, and oligosaccharides) is of great significance in revealing the basic mechanism and biological function in life activities. Inspired by nature, scientists have constructed a variety of supramolecular systems based on non-covalent interactions to study isomerization.^{7–12} However, given the diversity of structural combinations and dynamic processes in self-assembly, the construction and characterization of isomers at the supramolecular level remain great challenges. It is well-established that dimers are the simplest and most fundamental structural units for studying intermolecular interactions, and provide research models for explaining the formation and functional mechanism of more complex aggregates.^{13,14} According to the level of dynamics and the nature of linkages, most dimers can be classified into three categories: (i) dynamic non-covalent dimers, (ii) pseudo-static non-covalent dimers, and (iii) static covalent dimers.¹⁵ To date, many functional dimer systems exhibit significant application potential in various fields, including sensing,^{16,17} photochemistry,^{18,19} and catalysis.^{20,21} Nevertheless, there are still some great challenges in the structural design of dimers,

such as precise control of spatial distribution and orientation of groups, as well as fewer synthetic steps.

2,2':6',2''-Terpyridine (tpy) and its derivatives have been widely utilized to construct supramolecular architectures, due to their excellent coordination ability and stability.^{22,23} Tpy has many modifiable sites, especially at the pyridine rings on both sides, which provide versatile building blocks to construct a vast variety of complex architectures, including catenanes,²⁴ knots,²⁵ polygons,²⁶ and polyhedra.²⁷ Benefiting from the characteristic claw-like geometry, the 6-position modification of tpy enables the construction of supramolecular architectures with shape-complementary ligand pairs.²⁸ We have previously demonstrated that shape-complementary systems could provide a constrained environment to restrict the vibration or rotation of chromophores at the molecular level, thereby holding molecules in specific stereochemical conformations.²⁹ Inspired by this phenomenon, we further consider whether dimer isomerization would occur if dissymmetric substituent groups are introduced, which is similar to the different conformations of [2.2]paracyclophanes (PCPs) (Scheme 1a). A PCP consists of two cofacially stacked benzene rings with an average distance of 3.09 Å.³⁰ Substitution either at one ring or at both rings leads to different stereoisomers of PCP induced by the strained organic scaffold.³¹ Such dimer isomerization at the molecular level is expected to increase the complexity and diversity of the whole structure and offer new insight for designing stereoisomers.

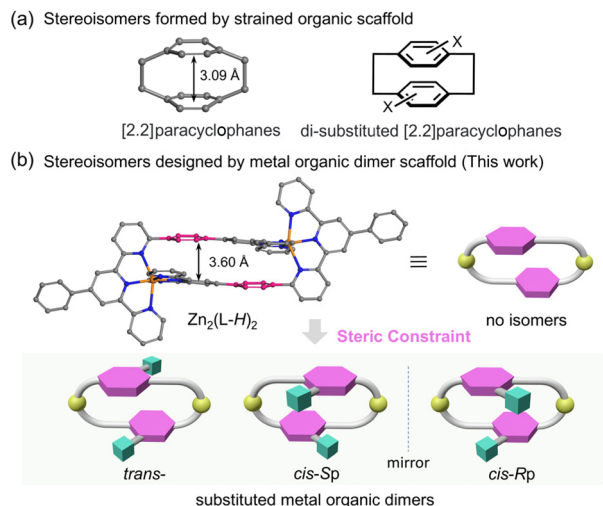
In the present work, we constructed a series of metal–organic dimers with different dissymmetrically modified substituents through the head-to-tail coordination mode of didentate tpy ligands. Owing to the serious steric constraint, the rotation of a substituted phenyl group was restricted, leading to different distributions and orientations of substituents (Scheme 1b). Single-crystal X-ray diffraction data demonstrated that the alkoxy substituents adopt a *cis* arrangement while

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Scheme 1 (a) Structural parameters of [2.2]paracyclophanes and disubstituted [2.2]paracyclophanes obtained by X-ray diffraction analysis; (b) X-ray crystal structure of $\text{Zn}_2(\text{L}-\text{H})_2$ and representative examples of the *trans*- or *cis*-arrangements of substituents induced by the steric constraint.

fluorine atoms adopt a *trans* arrangement in the crystal structure. More interestingly, these dimers show a strong tendency to pack due to the non-coaxial backbone of the structure, and the relationship between substituents and intermolecular interactions is investigated. This work not only provides a viable approach for designing stereoisomers, but also offers potential structure models for photochemistry and molecular devices.

For the synthesis of different di-dentate ligands **L-R**, the key precursor with dissymmetric modification can be efficiently obtained through the Kröhnke and Claisen–Schmidt condensation in decent yield (Fig. 1). After that, ligands **L-R** were obtained in 70%–82% isolated yields using the Suzuki coupling reaction. Complexes **Zn₂(L-R)₂** were obtained by assembling different ligands with Zn(NO₃)₂·6H₂O in an exact stoi-

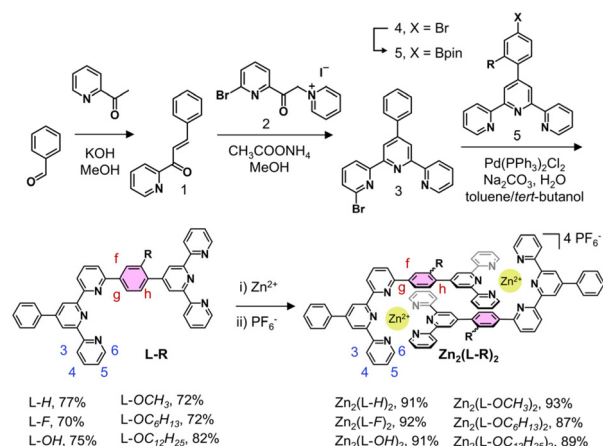


Fig. 1 Synthesis of ligands L-R and complexes $\text{Zn}_2(\text{L-R})_2$.

chiometric ratio of 1:1 in CHCl_3 and MeOH (v/v = 1:3) for 12 h at 50 °C, followed by the counterion exchange from NO_3^- to PF_6^- .

All ligands and complexes were fully characterized by ^1H and ^{13}C nuclear magnetic resonance (NMR) and two-dimensional correlation spectroscopy (2D COSY) (see ESI Fig. S5–S40†). In the aromatic region of the ^1H NMR spectrum, five sets of pyridine signals of **L-R** were observed due to the dissymmetric chemical environment induced by dissymmetric modification (Fig. 2). The 6-tpy protons showed apparently upfield shifts after coordination when compared with the free ligands, as expected. More importantly, for all dimeric complexes, protons H^f , H^g , and H^h in the middle phenyl group exhibit a dramatic upfield shift relative to the corresponding peaks for the free ligands as a result of a strong electronic shielding effect, *e.g.*, $\Delta\delta(\text{H}^f) \approx 0.46$ ppm, $\Delta\delta(\text{H}^g) \approx 1.39$ ppm, and $\Delta\delta(\text{H}^h) \approx 1.23$ ppm in **Zn₂(L-F)₂**, respectively. It was noted that the proton signals that belonged to tail tpy in complexes **Zn₂(L-R)₂** (R = F, OH, OCH₃, OC₆H₁₃, and OC₁₂H₂₅) became much more broad compared with those in **Zn₂(L-H)₂**, suggesting a seriously restricted chemical environment. In the high-resolution electrospray ionization mass spectrometry (ESI-MS) spectrum, all complexes can be unambiguously identified by the charge states from +2 to +4 due to the loss of a different number of PF_6^- (Fig. S45†). After deconvolution, the molecular weights of the complexes matched well with the expected chemical compositions of the two ligands, two Zn(II) ions, and four PF_6^- . The isotope pattern of each peak agreed well with the corresponding simulated isotope pattern (Fig. S46†). Combined with the results of NMR and ESI-MS, it can be concluded that different substituents do not have an

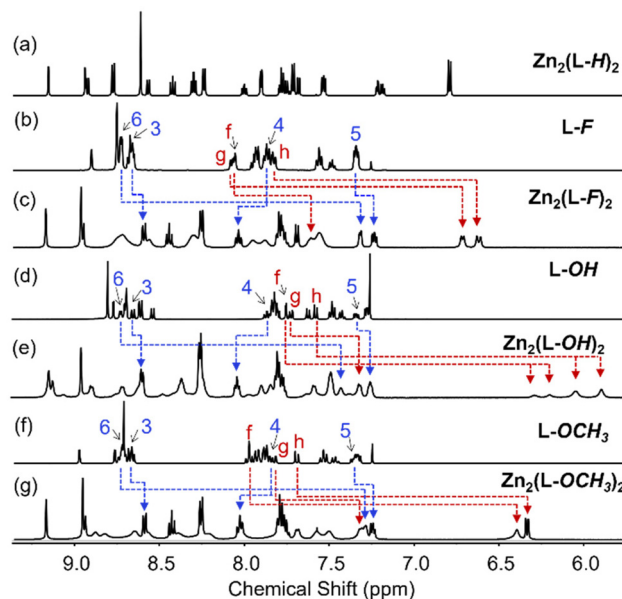


Fig. 2 ^1H NMR (500 MHz, 298 K) spectra of ligands (b) L-F, (d) L-OH, and (f) L-OCH₃ in CDCl₃, and complexes (a) Zn₂(L-H)₂, (c) Zn₂(L-F)₂, (e) Zn₂(L-OH)₂, and (g) Zn₂(L-OCH₃)₂ in CD₃CN.

effect on the formation of dimer complexes with a head-to-tail structure, even if the substituents exhibit bulky size.

For crystallographic investigations, single crystals of these complexes were successfully obtained by slowly diffusing the vapor of a poor solvent into an acetonitrile solution of the complexes at 15 °C for three or four weeks (Fig. S47–S55, and Tables S1–S4†). As we expected, all complexes $\text{Zn}_2(\text{L-R})_2$ showed a head-to-tail coordination structure, in which two Zn (II) ions are coordinated by two shape-complementary ligands. The centroid–centroid distances between the substituted phenyl group and pyridine for $\text{Zn}_2(\text{L-H})_2$, $\text{Zn}_2(\text{L-F})_2$, and $\text{Zn}_2(\text{L-OH})_2$ are 3.60 Å, 3.59 Å, and 3.53 Å, respectively. When the substituents were symmetric, such as **L-H**, the dimer had no isomers in the crystalline state (Fig. 3a). However, when the

substituents were replaced using the dissymmetric approach, different stereoisomers were generated in the crystalline state. Complex $\text{Zn}_2(\text{L-F})_2$ shows a similar structure to that of $\text{Zn}_2(\text{L-H})_2$, but two -F substituents in $\text{Zn}_2(\text{L-F})_2$ are located on the opposite side of the backbone in a *trans* conformation (Fig. 3b). Interestingly, for the dissymmetric substituents of $\text{Zn}_2(\text{L-OH})_2$, two hydroxyl groups adopt a *cis* arrangement close to each other and create intramolecular hydrogen bonding (Fig. 3c). The two different substituents also lead to different packing structures of the complexes. The crystal structure of $\text{Zn}_2(\text{L-F})_2$ (orthorhombic, space group *Pbca*) adopted a fish-bone-like stacking of dimers along the *b* axis due to the π – π interaction between tpy moieties (Fig. 3d). For the $\text{Zn}_2(\text{L-OH})_2$ complex, the *cis* arrangement of the hydroxyl group generated two stereoisomers of $\text{Zn}_2(\text{L-OH})_2$, e.g., *cis-Sp-Zn* $_2(\text{L-OH})_2$ and *cis-Rp-Zn* $_2(\text{L-OH})_2$, which are enantiomers of each other. The racemic mixture of $\text{Zn}_2(\text{L-OH})_2$ crystallized in the orthorhombic space group *Ccce* (Fig. 3e). One unit cell consists of eight dimers with four *cis-Sp-Zn* $_2(\text{L-OH})_2$ units and four *cis-Rp-Zn* $_2(\text{L-OH})_2$ units, respectively. Two complexes of homochirality interact with each other through intermolecular hydrogen bonding. It is worth mentioning that $\text{Zn}_2(\text{L-OH})_2$ was racemic in solution due to the dynamic nature of the coordination bond despite the hydrogen bond between two oxhydryl groups.^{32,33}

For the dissymmetric substituents of complexes $\text{Zn}_2(\text{L-R})_2$ (R = OCH₃, OC₆H₁₃, and OC₁₂H₂₅), all alkoxy groups adopted a *cis* arrangement and the backbones of these complexes became distorted to accommodate the arrangement of alkyl chains with a bulky size (Fig. 4a–c). The dihedral angles between the middle benzene plane and pyridine of $\text{Zn}_2(\text{L-OCH}_3)_2$, $\text{Zn}_2(\text{L-OC}_6\text{H}_{13})_2$, and $\text{Zn}_2(\text{L-OC}_{12}\text{H}_{25})_2$ are 42.43°, 32.41°, and 38.79°, respectively. The centroid–centroid distances between the phenyl group and pyridine are 3.60 Å, 3.53 Å, and 3.40 Å, respectively. The crystal structures of $\text{Zn}_2(\text{L-OCH}_3)_2$ and $\text{Zn}_2(\text{L-OC}_{12}\text{H}_{25})_2$ exist within a triclinic *P* $\bar{1}$ space group, while the crystal structures of $\text{Zn}_2(\text{L-OC}_6\text{H}_{13})_2$ exist within a monoclinic *P*₂₁/*c* space group. Two enantiomers, *cis-Sp-Zn* $_2(\text{L-OCH}_3)_2$ and *cis-Rp-Zn* $_2(\text{L-OCH}_3)_2$ were alternately stacked along the *c* axis (Fig. 4d). $\text{Zn}_2(\text{L-OC}_6\text{H}_{13})_2$ and $\text{Zn}_2(\text{L-OC}_{12}\text{H}_{25})_2$ both adopted parallel stacking of enantiomers due to the CH $\cdots\pi$ interaction between the alkyl chain and phenyl group (Fig. 4e).

By introducing different substituents in complexes $\text{Zn}_2(\text{L-R})_2$, stereoisomers of dimers in the crystal state were created. All substituents adopted a *cis* arrangement except for fluoro substituents, indicating that the size of the substituents was not the only factor in deciding whether the substituents adopted a *trans* or *cis* arrangement. To further understand the quite different arrangement of fluoro substituents, density functional theory (DFT) calculations were employed to calculate the energy of the isomers of $\text{Zn}_2(\text{L-F})_2$ (Fig. 5). The calculated relative energy of the *cis-Rp-Zn* $_2(\text{L-F})_2$ isomer was much higher (10.61 kJ mol^{−1}) than that of *trans-Zn* $_2(\text{L-F})_2$ (3.86 kJ mol^{−1}). From the result of DFT calculations, $\text{Zn}_2(\text{L-F})_2$ preferred the *trans* conformation for crystal stacking while it was

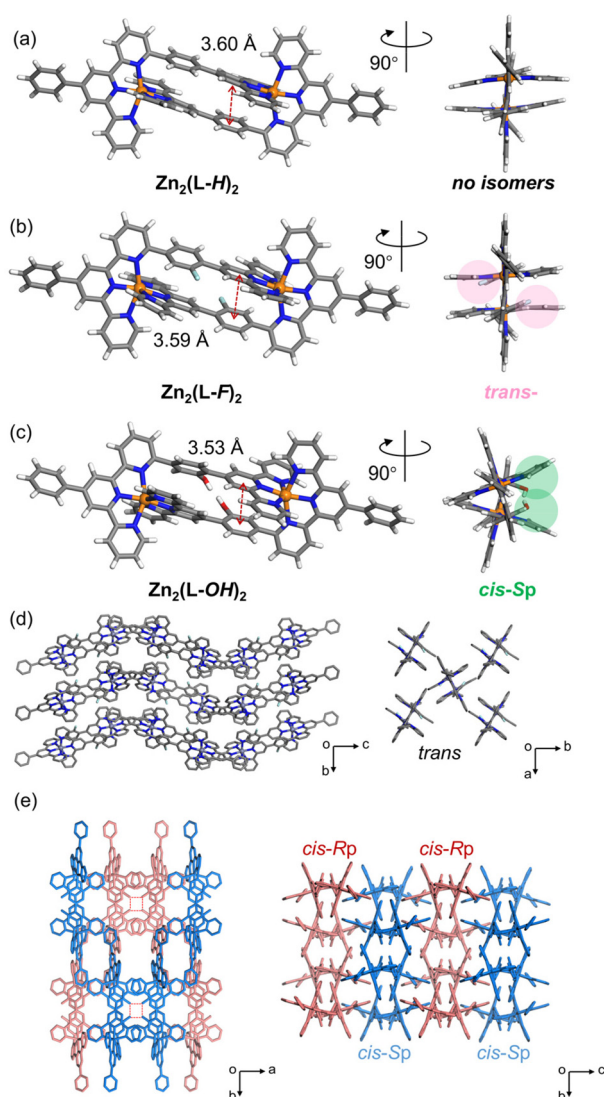


Fig. 3 X-ray crystal structures of (a) $\text{Zn}_2(\text{L-H})_2$, (b) $\text{Zn}_2(\text{L-F})_2$, and (c) $\text{Zn}_2(\text{L-OH})_2$ in the front and side views. Crystal packing structures of (d) $\text{Zn}_2(\text{L-F})_2$ and (e) $\text{Zn}_2(\text{L-OH})_2$. Hydrogen atoms, non-coordinated anions, and solvents are omitted for the sake of clarity.

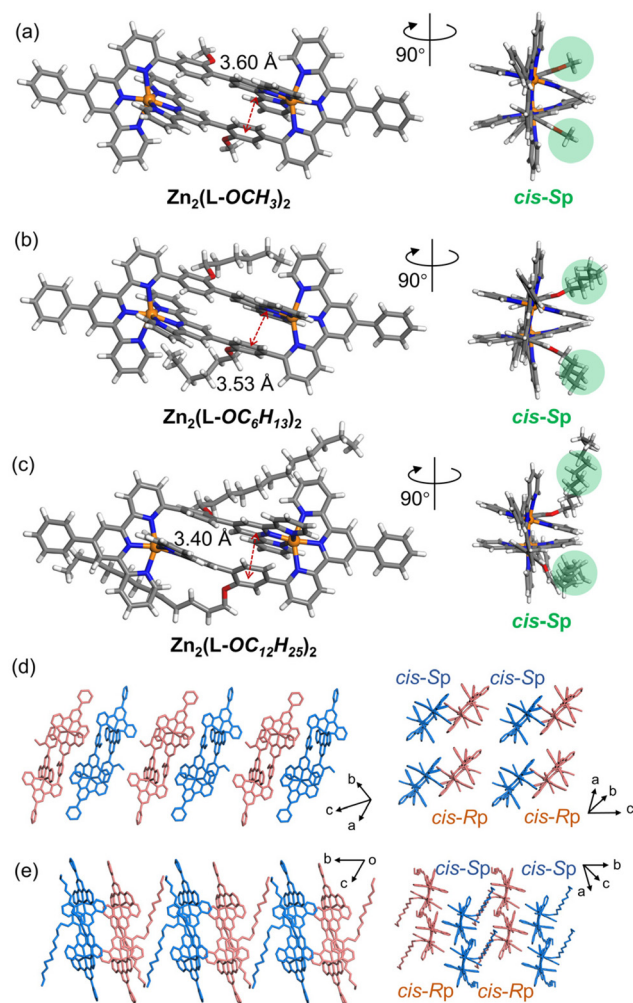


Fig. 4 X-ray crystal structures of (a) $\text{Zn}_2(\text{L-OCH}_3)_2$, (b) $\text{Zn}_2(\text{L-OC}_6\text{H}_{13})_2$, and (c) $\text{Zn}_2(\text{L-OC}_{12}\text{H}_{25})_2$ in the front and side views. Crystal packing structures of (d) $\text{Zn}_2(\text{L-OCH}_3)_2$ and (e) $\text{Zn}_2(\text{L-OC}_{12}\text{H}_{25})_2$. Hydrogen atoms, non-coordinated anions, and solvents are omitted for the sake of clarity.

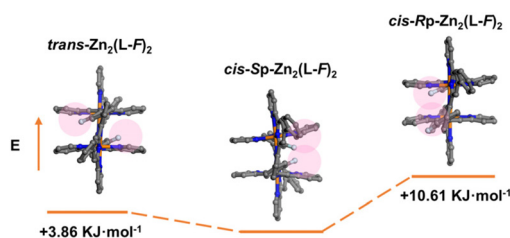


Fig. 5 Calculated structures (PBE1PBE/6-31G(d)) of the isomers of $\text{Zn}_2(\text{L-F})_2$ with the energies relative to the $\text{cis-Sp-Zn}_2(\text{L-F})_2$ isomer.

energetically unfavorable for the stacking of *cis* conformations. In addition, the repulsion between two electron clouds of a fluoro substituent might also be one of the factors leading to the *trans* conformation.

Conclusions

In summary, we constructed a series of metal–organic dimers with different stereochemical conformations based on *ortho*-modified terpyridine. X-ray crystallography confirmed that alkoxy substituents adopt a *cis* arrangement while fluoro substituents adopt a *trans* arrangement in the crystal structure. This result might be attributed to the lower energy of the *trans* arrangement of fluorine according to the DFT calculations. The substituents also introduce different driving forces for intermolecular packing and lead to different arrangements of complexes. These findings will offer new insight for the design and synthesis of stereoisomers and the further applications of metal–organic dimers.

Conflicts of interest

There are no conflicts to declare.

Data availability

The data supporting this article have been included as part of the ESI.†

Acknowledgements

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